

Chapter

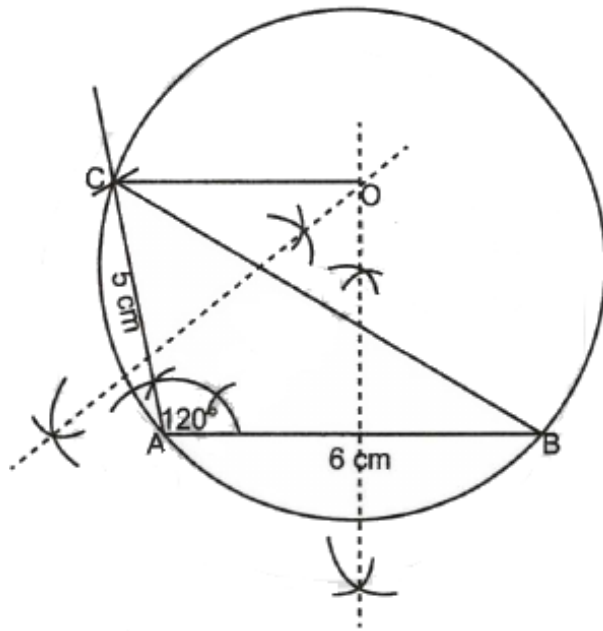
Constructions

1. Using ruler and compass construct a triangle ABC in which $AB = 6$ cm, $\angle BAC = 120^\circ$ and $AC = 5$ cm. Construct a circle passing through A, B, and C. Measure and write down the radius of the circle. Main 2023

2. Draw a circle of radius 4 cm. Draw two tangents to this circle so that the angle between the tangents is 60° . Comp 2023

Solution

1.



Steps of construction:

1. Draw $AB = 6$ cm.
2. At A, draw 120° and mark $AC = 5$ cm. Join BC.
3. Draw perpendicular bisectors of AB and AC; they meet at O.
4. Measure OC (radius).
5. Draw a circle with centre O and radius OC through A, B, and C.
6. Radius $OC = 5.5$ cm.

2.

Steps of Construction :

Step 1 : Draw a circle of radius 4cm with centre O

Step 2 : Draw two arcs OT and PT with $180^\circ - 60^\circ = 120^\circ$ between them.

Step 3 : At S and T draw rays making 90° each which meet each other at P .

Now $\angle TPS = 60^\circ$ and PT and PS are required tangents to the circle.

